

Shared Genetic Architecture between Bacterial Pneumonia and Cardiovascular Disease: a GWAS-eQTL Colocalization Analysis

Progress Report - Pneumonia pilot
Jaouad El Morabit (Data & QC Lead) | Team: C. Taha, L. Celistin, H. Arbaoui | Supervisor: Prof. Cherng Yang
16 July 2026

1. Objective and hypothesis

This project investigates whether bacterial infection susceptibility and cardiovascular disease (CVD) share a common genetic architecture. The central hypothesis is that infection-susceptibility and CVD share causal genetic variants acting through immune and inflammatory pathways, detectable as an overlap between GWAS signals and tissue-specific expression QTLs (eQTL). Pneumonia is the pilot trait. The analytical logic is: a pneumonia GWAS signal that colocalizes with a tissue eQTL implicates a candidate gene; we then test whether that gene belongs to a curated CVD gene list and interpret it biologically.

2. Data and methods

GWAS. FinnGen Data Freeze 13, endpoint J10_PNEUMONIA (European ancestry, GRCh38/hg38): 21.3M variants, 501 genome-wide significant ($p < 5e-8$). Colocalization parameters: $N = 500,186$; cases = 84,364; proportion of cases $s = 0.1687$; study type = case-control.

eQTL. GTEx v8 retrieved through the EBI eQTL Catalogue REST API v2 - whole blood (QTD000356, $n = 670$) and lung (QTD000271, $n = 510$).

Harmonization (three checks). SNP matching by position and sorted allele pair; effect-allele alignment (eQTL beta sign flipped when ref/alt are swapped relative to the GWAS); removal of ambiguous A/T and C/G SNPs near MAF 0.5. Both datasets are hg38, so no liftOver was required.

Colocalization. coloc.abf (beta/varbeta parameterization), run per gene per tissue. A locus is declared colocalized when the posterior probability of a shared causal variant PP.H4 exceeds 0.5 (threshold set by the supervisor).

3. GWAS scan - loci with signal

The full scan identified 501 genome-wide significant variants distributed across seven chromosomes:

Chr	Hits	Main genes	Used for coloc?
1	82	CRP, CRPP1	Yes (CRP locus)
2	1	IFIH1	No - not testable (see 5)
5	4	CDH6	No - underpowered
6	251	HLA-DQA1/DQB1, HCP5 (MHC)	No - excluded (extreme LD)
9	11	TNFSF15, DEC1	Yes
12	12	HNF1A	Yes
15	140	CHRNA3/5, HYKK, IREB2, PSMA4	Yes

4. Colocalization results

Four well-powered, non-MHC loci were taken through the full pipeline (eQTL fetch, harmonization, coloc.abf) in whole blood and lung:

Locus	GWAS hits	Key candidate genes (strong cis-eQTL)	Best PP.H4	Colo c?
-------	-----------	---------------------------------------	------------	---------

chr15 (15q25)	140	CTSH, IREB2, CIB2, RASGRF1	~0.02	No
chr1 (CRP)	82	SLAMF8, IGSF8, COPA, AIM2, MNDA	0.10	No
chr12 (HNF1A)	12	P2RX4, COQ5, C12orf43	0.29	No
chr9 (TNFSF15)	11	TNFSF15, ORM1	<0.01	No

Master result. Across the four loci and 71 gene-tissue tests, there is **no colocalization** (all PP.H4 < 0.5; maximum 0.29, LINC01089 in lung). Genes with strong GWAS *and* strong eQTL signals consistently return **PP.H3 approx 1**: both signals are real but driven by **distinct causal variants**. Notably, even TNFSF15 itself (the ideal hypothesis gene, with a strong blood eQTL) is H3. Full per-gene table: results/MASTER_coloc_results.tsv.

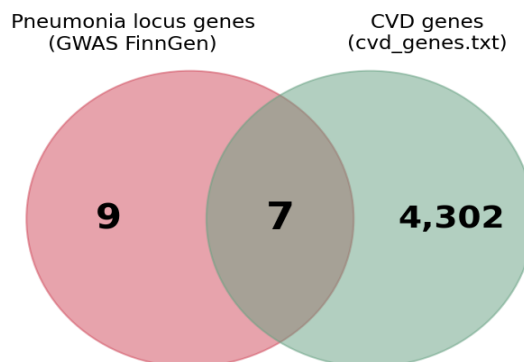
5. Loci that could not be tested

chr2 (IFIH1): only one genome-wide significant variant, and GTEx has no cis-eQTL for IFIH1 in blood or lung - colocalization is not possible. **chr5 (CDH6)**: too few hits (underpowered). **chr6 (MHC)**: excluded by design; the extreme, extended LD makes colocalization uninterpretable in this region.

6. Gene-level overlap with the CVD gene list (Phase 5)

Of the 16 genes located at pneumonia GWAS loci, **7 are also CVD genes** (cvd_genes.txt, 4,309 genes): **CHRNA3, CHRNA5, CRP, HCP5, HLA-DQB1, HNF1A, PSMA4** - including CRP and HNF1A, two direct inflammation-to-CVD links.

Pneumonia GWAS genes vs CVD genes - 7 / 16 shared



Shared (7): CHRNA3, CHRNA5, CRP, HCP5, HLA-DQB1, HNF1A, PSMA4

7. Interpretation

Pneumonia and CVD **share genomic loci and genes**, but colocalization finds **no evidence of shared causal variants** acting through cis-regulation of gene expression in whole blood or lung. In other words, the shared genetic architecture appears to sit at the level of overlapping regions and genes rather than identical causal variants - at least under the single-causal-variant assumption of coloc.abf, and in the two tissues examined. This is a consistent, reproducible result across all four testable loci rather than an isolated finding.

8. Open decisions - your validation requested

D1. Primary GWAS source. A second pneumonia GWAS (GCST90624053, GWAS Catalog) was evaluated. It is cleaner on chr15 but covers fewer loci (3 vs 7) and, importantly, **includes FinnGen R10**, so it is not independent from our FinnGen DF13 pilot. Our recommendation is to keep **FinnGen DF13** as the primary source for broader locus coverage. *Do you validate this choice?*

D2. coloc.susie. Because H3 is consistent at multi-signal loci (chr15, CRP, HNF1A), coloc.abf's one-causal-variant assumption may mask a hidden colocalization. coloc.susie would test this rigorously, but it requires an external LD reference (e.g. 1000 Genomes EUR) that we do not currently have set up. *Should we invest in running coloc.susie on one locus, or report the abf result as final with susie stated as a limitation?*

D3. Scope and tissues. The pilot used whole blood and lung only. Some lead genes (CRP, HNF1A) are primarily liver-expressed and have no blood/lung eQTL. *Do you want us to (a) freeze the pneumonia pilot as is, (b) add a disease-relevant tissue such as liver or artery, or (c) begin extending the validated pipeline to a second infection trait?*

9. Next steps and limitations

Next steps (pending your input): finalize D1-D3; complete the presentation; if approved, set up an LD reference for coloc.susie and/or extend to additional tissues or traits.

Limitations: (i) single-causal-variant assumption of coloc.abf at multi-signal loci; (ii) only two tissues examined; (iii) GTEx lacks cis-eQTLs for several lead genes (CRP, IFIH1, HNF1A), so their direct expression-mediated effects cannot be assessed here.